

Project notes: Empirical Market Microstructure

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Chapter 3: Roll model

Introduction:

- Introduction of random walk with non-zero mean and why it can be used as a model (stock price, log stock price) for stocks over long-term time horizons. Light discussion on why the model isn't fit for other securities like options for long-term but possible for short-term.
- Introduction of martingales and how they arise via assumptions that are not appropriate (e.g. frictionless trading). Random walk is a special example of a martingale.
- Distinction between fundamental value/efficient price of security and the transaction price (former is expectation of transaction price conditioned on previous prices)
- Something I found interesting: homogeneity vs. stationarity
- The book moves on to defending its assumptions on short-run security price characteristics: (1) near zero mean (2) extreme dispersion (3) dependence between successive observations

Near-zero mean:

- We are interested in modelling security prices using a random walk over short-term horizon.
- Over this horizon, the drift term has little time to accumulate; its contribution over a short-term interval is tiny relative to the random price variation.
- Over a longer time horizon, however, this drift accumulates and helps to describe the long-term trend (think about mean in a time series model).
- We set the mean to zero. It adds bias to the model but reduces the estimation error and also makes intuitive sense for the time horizon we are working over.

Extreme dispersion:

- Empirical price change distributions have fat tails, which begs the question of whether the true underlying distribution has finite kurtosis, skewness, or even variance.
- The important thing to understand is infinite moments cannot be directly observed using finite empirical data, hence the issue isn't with what we are observing, the issue is with how much we can trust the statistical conclusions drawn from the empirical data given that we know price change distributions may not have finite higher moments.
 - “How much can we trust the conclusions” analogy: given I sample 3 players from the NBA, and get an average height of 6'2, how much can I trust my conclusions knowing that the average nba player height is 6'7?
- Possible remedies: ad hoc transformations (downside is that if we are using price changes or log returns the resulting variable doesn't have as nice of an interpretation) or modelling the distribution as a mixture of well-behaved distribution (we can think about indicator variables and turning certain distributions on and off depending on the market regime; downside is that we don't directly observe the regime, this leads us to hidden states)

Dependence of successive changes:

- Introduce the notion of autocovariance and autocorrelation (see time series notes!)

Chapter 4: univariate time series

- Nothing really new besides ergodicity: local stochastic behaviour is independent of its starting point; an ergodic process “forgets” where it started.
- Roll model can be interpreted as a MA(1) process via Wold representation or equivalently AR(inf) (assuming invertibility).
- Predictions and forecasting can be done via conditional expectation. For simplicity (in solving the system and interpretability), we only look toward a linear predictor. If the variables are jointly Gaussian, the optimal predictor under mean-squared error is linear.
- Univariate time series models are great for building intuition behind how the model works and hence how the financial markets work under the model.
- Stationary models can be a useful local approximation, however, aren’t a realistic description of the market under all conditions.
- Additionally, knowing the first- and second-order properties of a time series does not generally uniquely specify its data-generating process.
 - Analogy: it’s like describing a person but it doesn’t narrow down who it is, so we aren’t able to conclusively say something about that certain person as the group of people all share specific properties.